Michael Theologitis

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PERSONAL STATEMENT

I study how to build AI systems that effectively use and manage context in long-horizon, data-intensive tasks. My work spans data representation, memory, and agent harnesses, extending to meta-harnesses and self-improvement. My broader goal is to enable people like journalists, lawyers, and scientists to answer complex questions over large collections of structured and unstructured data using smaller models they can run and control locally, while ensuring that the answers are trustworthy and inspectable.

EDUCATION

UNIVERSITY OF WASHINGTON | *PhD in Computer Science* 2025 – Present
- Advisors: Dan Suciu, Yulia Tsvetkov
- Interests: NLP, AI Agents, Data Management

TU-CRETE | *MS in Computer Science and Engineering* 2023 – 2025
- Supervised Fine-Tuning, LLMs, Federated Learning

TU-CRETE | *BEng & MEng in Electrical and Computer Engineering* 2017 – 2023
- Deep Learning, Convex Optimization, Linear Algebra, Statistics

EXPERIENCE

CREXDATA Europe
Research Software Engineer 2023 – 2025
- Developed a novel algorithm, reducing costs by 10×–100× for Federated Learning jobs.
- Built a platform to deploy and monitor 200+ concurrent FL jobs, each with 1,000+ devices on HPC environments (e.g., Barcelona's MareNostrum supercomputer).
- Fine-tuned LMs on uncertain streaming data to monitor and alert first-responders for emergency scenarios (e.g., floods, fires, etc.) for the Ministry of Interior of Finland.

SELECT PUBLICATIONS

Preprint ***Michael Theologitis****, ..., *Benjamin Newman, Yulia Tsvetkov, Dan Suciu* Realize What Matters: Principled Context Representation for Large-Scale Reasoning

NeurIPS '26 *Dean Light**, ***Michael Theologitis****, ..., *Pang Wei Koh, Dan Suciu, Yulia Tsvetkov* Deep Reasoning in General Purpose Agents via Structured Meta-Cognition

ACL '26 ★ Top 15% ***Michael Theologitis****, Preetam Dammu, Chirag Shah, Dan Suciu* ClaimDB: A Fact Verification Benchmark over Large Structured Data

AAAI '26 ★ Oral ***Michael Theologitis****, Dan Suciu* Thucy: An LLM-based Multi-Agent System for Claim Verification across Relational Databases *LaMAS'26 Workshop*

CIKM '26 ***Michael Theologitis****, Vasilis Samoladas, Antonios Deligiannakis* FDA-Opt: Communication-Efficient Federated Fine-Tuning of Language Models

***** Equal contribution

TECHNICAL SKILLS

Programming Python (10+ years), SQL (5+ years), Claude Code, C, Java, MATLAB

Agentic Frameworks Smolagents, PydanticAI, OpenAI Agents SDK, OpenHands

AI/ML HuggingFace, vLLM, PyTorch, TensorFlow, NumPy, Ollama, Pandas, Matplotlib, Seaborn

Databases PostgreSQL, MySQL, SQLite, MongoDB

Parallel Computing Spark, Dask, Hadoop, Kafka, HDFS, Yarn, Slurm, Docker

Tools & Platforms Linux, GitHub, Git, PyCharm, VS Code, JupyterLab, LaTeX, Powerpoint

SELECT PROJECTS

R^{3}CON: EFFECTIVE CONTEXT REPRESENTATION | *Link* Jan. 2026 – Present

- Developed R^{3}CON, a harness that helps LLMs **piece together interdependent evidence** scattered across large document corpora to solve tasks beyond their context limits.
- Translated the cognitive theory of relevance realization into **design principles for building AI systems that effectively organize large contexts for reasoning**; analyzed how existing systems' successes and failures relate to these principles.
- Evaluated against 9 state-of-the-art AI systems, improving accuracy over the strongest baseline by **20 and 8.4 percentage points** on CorpusQA and Loong, respectively.
- R^{3}CON enables smaller models to outperform much larger ones, with 4B and 9B models surpassing all evaluated 35B baselines on Loong and CorpusQA, respectively. Moreover, Qwen3.5-35B-A3B with **R^{3}CON outperforms Claude Code equipped with Claude-Sonnet-5** on CorpusQA at **3.7× lower cost**.

INSTILLING HUMAN META-REASONING IN AGENTS | *Link* Nov. 2025 – Feb. 2026

- Built a prompting strategy to take **human meta-reasoning** and convert it **into in-context examples** that agents can understand and deploy.
- Instilled the flexibility of human reasoning in general-purpose agents.
- Introduced the **DOLORES** agent and evaluated it across four hard benchmarks (e.g., long-horizon, multi-hop, deep search) improving over SOTA by **+24.8% accuracy** on average.

CLAIMDB: BENCHMARK FOR AGENTS | *Project Site* Nov. 2025 – Feb. 2026

- Benchmark for fact-verification over large relational databases: **80 DBs** and **53k facts**
- **For each fact**, it requires that agents explore structured data that is **+110M tokens** large
- Evaluated **30 state-of-the-art LLMs**, and showed models struggle in reasoning over large structured data: they **struggle in *abstention*** — the ability to admit *"I don't know"*.

THUCY: MULTI-AGENT SYSTEM FOR FACT VERIFICATION | *Link* Sept. 2025 – Nov. 2025

- Explores databases to uncover whether *facts* are supported from the structured data.
- Improved over the SOTA on the TabFact benchmark by **+5.6 percentage points**.

FDA-OPT: A FEDERATED OPTIMIZATION ALGORITHM | *Link* Sept. 2024 – Aug. 2025

- FDA-Opt completely eliminated the hard-to-calibrate communication interval τ from the family of optimization algorithms FedOpt (e.g., FedAvg, FedAdam, etc.).
- Fine-tuned LMs (encoders) on various NLP tasks and found that **FDA-Opt converges better than FedOpt**—across all individual optimizers like Adam, AdamW, SGD, SGDM, etc.—and is also around **2× more efficient**.

INVITED TALKS

RelationalAI Fact Verification over Structured Data: Agentic Systems and Benchmarks 2026

COMMUNICATION

Languages English (Proficient), French (Beginner), Greek (Native)

Presenting Delivered 50+ presentations of highly technical material to audiences of 40+ academic and industry partners, with a strong track record of creating clear, engaging, and effective slides for expert and non-expert audiences alike

Writing Strong academic and research writing skills, demonstrated through multiple first-author publications